

Written HW 2 — Full Solutions

Problem statements from Knight, *Physics for Scientists and Engineers* (5th ed., Global Edition), Chapter 22. $K = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$ and $g = 9.8 \text{ m/s}^2$ throughout.

22.17 (3 pts)

Three charges lie on the x-axis (Figure EX22.17): $q_A = +1.0 \text{ nC}$ at $x = 0$, $q_B = -1.0 \text{ nC}$ at $x = 1.0 \text{ cm}$, and $q_C = +4.0 \text{ nC}$ at $x = 2.0 \text{ cm}$. What is the magnitude of the net force on charge A?

MODEL: Treat A, B, and C as point charges. The net force on A is the vector sum of the force from B alone and the force from C alone.

VISUALIZE: B is negative, so it attracts A — that force pulls A to the right, toward B ($+\hat{x}$). C is positive, so it repels A — that force pushes A to the left, away from C ($-\hat{x}$). The two forces point in opposite directions.

SOLVE: By Coulomb's law,

$$F_{B \text{ on } A} = K|q_A q_B|/r_{AB}^2 = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.0 \times 10^{-9} \text{ C})(1.0 \times 10^{-9} \text{ C})/(0.010 \text{ m})^2 = 8.99 \times 10^{-5} \text{ N, direction } +\hat{x}$$

$$F_{C \text{ on } A} = K|q_A q_C|/r_{AC}^2 = (8.99 \times 10^9)(1.0 \times 10^{-9})(4.0 \times 10^{-9})/(0.020 \text{ m})^2 = 8.99 \times 10^{-5} \text{ N, direction } -\hat{x}$$

These are equal in magnitude and opposite in direction, so

$$F_{\text{net},A} = (8.99 \times 10^{-5} \text{ N})\hat{x} + (8.99 \times 10^{-5} \text{ N})(-\hat{x}) = \mathbf{0 \text{ N}}$$

REVIEW: This cancellation isn't an accident — C sits twice as far from A as B does, but carries four times the charge, and Coulomb's law goes as (charge)/(distance)². Doubling the distance cuts the force by 4×, and quadrupling the charge multiplies it by 4×, so the two effects exactly offset. A good example of how sensitive the inverse-square law is to distance.

22.18 (3 pts)

Three charges lie on a vertical line (Figure EX22.18): $q_A = +1.0 \text{ nC}$ at the top, $q_B = -2.0 \text{ nC}$ located 2.0 cm below A, and $q_C = +2.0 \text{ nC}$ located 1.0 cm below B. What is the magnitude of the net force on charge B?

MODEL: Treat A, B, and C as point charges.

SOLVE: A is positive and B is negative, so A attracts B upward, toward A:

$$F_{A \text{ on } B} = K|q_A q_B|/r_{AB}^2 = (8.99 \times 10^9)(1.0 \times 10^{-9})(2.0 \times 10^{-9})/(0.020 \text{ m})^2 = 4.5 \times 10^{-5} \text{ N, direction } +\hat{j} \text{ (up)}$$

C is positive and B is negative, so C attracts B downward, toward C:

$$F_{C \text{ on } B} = K|q_C q_B|/r_{CB}^2 = (8.99 \times 10^9)(2.0 \times 10^{-9})(2.0 \times 10^{-9})/(0.010 \text{ m})^2 = 3.6 \times 10^{-4} \text{ N, direction } -\hat{j} \text{ (down)}$$

Adding these as vectors,

$$F_{\text{net},B} = (4.5 \times 10^{-5} \text{ N})\hat{j} - (3.6 \times 10^{-4} \text{ N})\hat{j} = -3.1 \times 10^{-4} \text{ N } \hat{j}, \text{ so the magnitude is } \mathbf{3.1 \times 10^{-4} \text{ N}} \text{ (downward)}$$

REVIEW: C is closer to B than A is (1.0 cm vs. 2.0 cm) and carries twice A's charge, so it makes sense that C's pull wins decisively — unlike Problem 22.17, nothing here is set up to cancel.

22.19 (3 pts)

Three charges sit at the corners of an equilateral triangle with 1.0 cm sides (Figure EX22.19): $q_A = +2.0$ nC at the top vertex, $q_B = -2.0$ nC at the bottom-left vertex, and $q_C = +1.0$ nC at the bottom-right vertex. What is the force $\vec{F}$ on the 1.0 nC charge? Give your answer as a magnitude and direction.

MODEL: Treat all three as point charges. Charge C feels two forces — a repulsion from A and an attraction to B — that must be added as vectors.

VISUALIZE: Put B at the origin and C at (1.0 cm, 0), so A sits at (0.5 cm, 0.87 cm), completing the equilateral triangle. A and C are both positive, so A repels C straight along the line from A through C. B and C are opposite in sign, so B attracts C straight along the line from C to B.

SOLVE: Both pairs are separated by the same 1.0 cm side, and both charge products have the same magnitude ($2.0 \text{ nC} \times 1.0 \text{ nC}$), so the two individual forces come out equal:

$$F = Kq_Aq_C/r^2 = (8.99 \times 10^9)(2.0 \times 10^{-9})(1.0 \times 10^{-9})/(0.010 \text{ m})^2 = 1.80 \times 10^{-4} \text{ N}$$

The unit vector from A to C is (0.50, -0.87), so the repulsion from A contributes

$$\vec{F}_{A \text{ on } C} = (1.80 \times 10^{-4} \text{ N})(0.50, -0.87) = (9.0 \times 10^{-5}, -1.56 \times 10^{-4}) \text{ N}$$

The unit vector from C to B is (-1, 0), so the attraction to B contributes

$$\vec{F}_{B \text{ on } C} = (1.80 \times 10^{-4} \text{ N})(-1, 0) = (-1.80 \times 10^{-4}, 0) \text{ N}$$

Adding components,

$$\vec{F}_{\text{net},C} = (-9.0 \times 10^{-5}, -1.56 \times 10^{-4}) \text{ N} \Rightarrow \text{magnitude} = \sqrt{[(9.0 \times 10^{-5})^2 + (1.56 \times 10^{-4})^2]} = \mathbf{1.8 \times 10^{-4} \text{ N}}$$

Both components are negative, so $\vec{F}$ points into the third quadrant, at $\tan^{-1}(1.56/0.90) = \mathbf{60^\circ \text{ below the } -x\text{-axis}}$ — i.e., straight out through C, away from the midpoint of edge AB.

REVIEW: Because the two individual forces are equal in magnitude and 120° apart, the resultant must bisect that angle — which is exactly what the components give. That the two forces come out equal at all is itself a direct consequence of q_Aq_C and $|q_B|q_C$ being the same product (2.0 nC^2) at the same 1.0 cm separation.

22.33 (2 pts)

A +12 nC charge is located at the origin. What are the electric fields at the positions $(x, y) = (5.0 \text{ cm}, 0.0 \text{ cm})$, $(-5.0 \text{ cm}, 5.0 \text{ cm})$, and $(-5.0 \text{ cm}, -5.0 \text{ cm})$? Write each electric field vector in component form.

MODEL: The field of a positive point charge points radially outward from it, with magnitude $E = Kq/r^2$.

SOLVE: At $(5.0 \text{ cm}, 0)$: $r = 5.0 \text{ cm}$, straight along $+\hat{x}$.

$$E = (8.99 \times 10^9)(12 \times 10^{-9})/(0.050 \text{ m})^2 = 4.3 \times 10^4 \text{ N/C} \Rightarrow \vec{E} = (4.3 \times 10^4, 0) \text{ N/C}$$

At $(-5.0 \text{ cm}, 5.0 \text{ cm})$: $r = \sqrt{50} \text{ cm} = 7.07 \text{ cm}$, direction $(-1/\sqrt{2}, 1/\sqrt{2})$.

$$E = (8.99 \times 10^9)(12 \times 10^{-9})/(0.0707 \text{ m})^2 = 2.16 \times 10^4 \text{ N/C} \Rightarrow \vec{E} = (-1.5 \times 10^4, 1.5 \times 10^4) \text{ N/C}$$

At $(-5.0 \text{ cm}, -5.0 \text{ cm})$: same radius as the point above, direction $(-1/\sqrt{2}, -1/\sqrt{2})$, so

$$\vec{E} = (-1.5 \times 10^4, -1.5 \times 10^4) \text{ N/C}$$

REVIEW: The last two points are mirror images of each other across the x-axis, at the same distance from the charge, so their field magnitudes should match — they do ($2.16 \times 10^4 \text{ N/C}$ both) — and only the sign of the y-component flips, exactly as the geometry demands.

22.34 (4 pts) BIO

A 0.10 g honeybee acquires a charge of +23 pC while flying.

- The earth's electric field near the surface is typically 100 N/C, downward. What is the ratio of the electric force on the bee to the bee's weight?
- What electric field (strength and direction) would allow the bee to hang suspended in the air?

MODEL: Treat the bee as a point charge in a uniform field. Its weight is $F_g = mg$; the electric force on it is $F_e = qE$.

SOLVE: (a) The ratio of the two forces is

$$F_e/F_g = qE/mg = (23 \times 10^{-12} \text{ C})(100 \text{ N/C}) / [(0.10 \times 10^{-3} \text{ kg})(9.8 \text{ m/s}^2)] = \mathbf{2.3 \times 10^{-6}}$$

(b) For the electric force to balance gravity, $qE = mg$, so

$$E = mg/q = (0.10 \times 10^{-3} \text{ kg})(9.8 \text{ m/s}^2) / (23 \times 10^{-12} \text{ C}) = \mathbf{4.3 \times 10^5 \text{ N/C, directed upward}}$$

(so the force on the positive charge points up, opposing gravity).

REVIEW: Part (a)'s ratio is minuscule — Earth's ordinary fair-weather field is millions of times too weak to matter to a flying bee. Part (b)'s answer, $4.3 \times 10^5 \text{ N/C}$, is roughly 1000× the field strength that causes dielectric breakdown (sparking) in air, a reminder that electrically suspending a bee like this isn't something you could actually set up in a room, even though the algebra is clean.

22.36 (3 pts)

A +5.0 nC charge sits at the origin (Figure EX22.36). Point 1 is at (x, y) = (1.0 cm, 2.0 cm), point 2 is at (1.0 cm, 0 cm), and point 3 is at (1.0 cm, -2.0 cm). What are the electric fields at points 1, 2, and 3? Give your answer in component form.

MODEL: The field of a positive point charge points radially outward from it, with magnitude $E = Kq/r^2$.

SOLVE: At point 2: $r = 1.0$ cm, straight along $+\hat{x}$.

$$E = (8.99 \times 10^9)(5.0 \times 10^{-9})/(0.010 \text{ m})^2 = 4.5 \times 10^5 \text{ N/C} \Rightarrow \vec{E}_2 = (4.5 \times 10^5, 0) \text{ N/C}$$

Points 1 and 3 are both $r = \sqrt{5}$ cm = 2.24 cm from the charge, with direction $(1/\sqrt{5}, \pm 2/\sqrt{5})$.

$$E = (8.99 \times 10^9)(5.0 \times 10^{-9})/(0.0224 \text{ m})^2 = 8.99 \times 10^4 \text{ N/C}$$

$$\vec{E}_1 = (4.0 \times 10^4, 8.0 \times 10^4) \text{ N/C} \quad \vec{E}_3 = (4.0 \times 10^4, -8.0 \times 10^4) \text{ N/C}$$

REVIEW: Point 2 is closer to the charge (1.0 cm) than points 1 and 3 (2.24 cm), so it should have the larger field — and by a factor of about $(2.24)^2 \approx 5$, exactly what the numbers show (4.5×10^5 vs. 8.99×10^4 N/C). Points 1 and 3 are mirror images across the x-axis, so their fields are mirror images too, differing only in the sign of the y-component.